WHEN THERE IS NO RIGHT ANSWER: EVALUATION FAILURE AND A SAMPLING-SCALE FIX FOR LOW-RESOLUTION PIXEL-ART TEXTURES

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ABSTRACT

Low-resolution pixel-art texture generation exposes a blind spot in how generative models are evaluated. When the output is a 16×16 grid of discrete colour cells, independent artists drawing the *same* material agree on 9.8% of cells against 8.6% expected under a paired null (50,753 artist pairs) — there is no unique right answer. We show this single fact breaks both mainstream automatic evaluation routes. Per-pixel metrics measure marginal fit rather than quality. And VLM judges fail in ways that neither standard judge diagnostic detects: on the same human-labelled set, two judges agree with the human on *identical* fractions of items (66.7%), yet one compresses an 87% effect to 62% (p=0.14) and erases its stratification while the other preserves both; a third inverts the human conclusion outright. Self-consistency is no better a guide — the judge that erases the stratification is the most consistent of the three under order swapping. Under human blind comparison a trivial downsampling baseline beats a purpose-built masked-prediction model 83:17, and is indistinguishable from artist-drawn tiles under both arbiters (43% by a human, 45% by the validated judge). The bottleneck is not model capacity but a *structural-unit convention* mismatch: the generator draws $\sim$25 units per tile regardless of canvas while artists draw $\sim$3.2, measured independently from 5,979 artist tiles and 125 artist sources. Cropping to the detected dominant period repairs this on the 45–58% of prompts where a period is found, and manipulating the unit count produces a monotone dose–response in the fix's benefit (88%, 54%, 17% at 27.7, 13.8 and 9.4 units). On the rest, where artists draw per-cell colour rather than structural units, we close three candidate routes and find none that works. We pre-registered and *falsified* a pure sampling-limit account along the way; reporting that failure is what selects the surviving mechanism.

1 INTRODUCTION

The task sounds small. Given a solid-colour region and a material word — “brick”, “wood” — produce a pixel-art texture for it at 16×16, 24×24 or 32×32. But shrinking the canvas does something a larger canvas hides: it makes the space of acceptable answers large relative to the space of possible ones, and, unusually, it makes the degree of non-uniqueness *exactly computable*. A tile is a small grid of discrete palette indices, so two artists’ renditions of the same material either place the same index in a cell or they do not, with nothing to interpolate and no perceptual metric standing in between.

Measured that way, independent artists agree on 9.8% of cells against 8.6% expected under a matched null (§3.1). Whatever a per-pixel loss is fitting on this task, it is not something artists agree about. This single fact is the paper’s hinge, and it disqualifies both of the routes normally taken to evaluate a generative model here.

Both automatic evaluation routes fail, for the same reason. Reference-based quantities have almost no signal to work with, so they reduce to measuring marginal fit. We watched the consequence before we understood it: across three rounds of architecture changes, validation accuracy climbed while the sample galleries did not visibly change (§3). The other route, a VLM judge, fails differently and in a way the literature’s standard diagnostic cannot see. Two frontier judges agreed with human labels on *exactly* the same fraction of items, 66.7%, and one preserved the human conclusion while the other destroyed it (§3.3). Per-item agreement is not what makes a judge usable; reproducing a conclusion humans have already established is.

Under valid evaluation, a trivial baseline wins. Once the arbiter is a human blind comparison, the ordering our automatic quantities had reported inverts. Plain downsampling of a suitable high-resolution source beats our own purpose-built masked-prediction model 83:17 in a pre-registered study, with the loss concentrated exactly where our hand-designed structural priors act (§4). Against artist-drawn tiles the same baseline is indistinguishable (43%, p=0.68). This overturns the premise the project began from — that low-resolution pixel art is an artist's redesign rather than a downsampling — and it means a method has to earn its place somewhere other than where we had been looking.

The bottleneck is a scale convention, not capacity. Where no suitable source exists it has to be generated, and there the baseline is weak — not because the generator cannot draw the material, since its 1024px render is structurally sound, but because it draws about 25 structural units per tile while pixel artists place about 3.2 (§5). Cropping the render to its detected dominant period repairs it. We pre-registered a pure sampling-limit explanation of why that works and *falsified* it; reporting the failure is what selects the surviving account, which then passed a second pre-registered test (§5.4).

Contributions.

- **Diagnosis.** An exactly computable inter-author agreement test, with the paired null it requires, showing the task has no unique reference, and evidence that per-pixel metrics and VLM judges both fail on it (§3).
- **Baseline calibration.** Pre-registered blind comparisons showing a trivial baseline beats a purpose-built method and matches artists (§4).
- **Mechanism and fix.** Measurement of the structural-unit convention mismatch, a period-based cropping rule with a data-derived applicability gate validated on held-out materials, pre-registered causal evidence with a dose–response, and a pre-registered replication on 60 materials chosen after the method was fixed (§5).
- **Evaluation methodology.** A judge-admissibility protocol: validate on conclusion reproduction rather than per-item agreement, and let the judge's measured bias direction determine which claims it can support (§3.3).

2 Related Work

Pixel-art generation. The closest work is SD-πXL (Binninger & Sorkine-Hornung, 2024), which produces palette-constrained low-resolution imagery from a text prompt, optionally guided by a reference image, via score distillation. We reproduced it; two properties matter here. It has no region-constraint channel, so it cannot be asked to paint only inside a given shape, and raising its conditioning strength degrades rather than improves the output. More recent palette-aware diffusion models report FID (Heusel et al., 2017), palette-consistency and prompt-alignment scores alongside human preference. Our §3 questions the first family of numbers on this task family: reference-based quantities presuppose a reference, and here the per-cell agreement between independent authors of the same material is 0.098 against a paired null of 0.086. There is no unique reference to measure distance from.

Structure-preserving downscaling. Kopf et al. (2013) optimise the shape and position of sampling kernels to align with local features, and Öztireli & Gross (2015) cast downscaling as optimisation against an SSIM (Wang et al., 2004) objective; Wu et al. (2022) learn cell-controllable pixelization directly. This line asks how to lose the least structure at a given scale factor. Our diagnosis sits upstream of it: when the source packs $\sim$25 structural units into a 16-cell output, no kernel can represent them, and we confirm that five downsamplers fail identically on the same renders (§5). The remedy has to act on the scale factor itself.

VLM-as-judge and its biases. Automatic judging with strong models is now standard (Zheng et al., 2023), and positional bias is a known failure (Wang et al., 2024). Recent benchmarks quantify it carefully: Li et al. (2026) report order-swap consistency of 83.5–89.6% across judges on web-development comparisons, with first-position preference between 0.9% and 15.8%, and argue the bias reflects a model deficiency rather than annotation ambiguity, since explicit instructions do not remove it; judges rank more reliably than they score (Kumar et al., 2026). On pixel-art comparisons we find no positional bias, markedly lower self-consistency than that range for two of three judges, and a distinct and more consequential failure (§3.3): judges that look equally good on both diagnostics differ completely in whether they reproduce the human *conclusion*. We argue this conclusion-level validation, not agreement with human labels item by item, is what licenses using a judge.

Evaluation when the reference is not unique. Independent redraws of the same subject — icon themes, emoji sets, texture packs — are a natural testbed for reference-scarce evaluation, but the degree of non-uniqueness is usually not measurable. A discrete palette on a small grid makes it exactly computable, and we contribute that test, together with the paired null it requires, as a task-level diagnostic to run *before* choosing a metric.

3 There Is No Right Answer

3.1 Independent artists barely agree on individual cells

A 16×16 tile drawn from a discrete palette makes agreement exactly computable: two renditions of the same material either place the same palette index in a cell or they do not. We collected 4,951 such tiles covering 517 materials from 64 independently authored texture packs, and compared every pair of tiles depicting the same material — 50,753 pairs.

The comparison needs a null, and the choice of null is where this measurement is easiest to get wrong. Comparing artist A's tile against a shuffle of *its own* cells has expectation $\sum_i p_A(i)^2$, whereas the real comparison A against B has expectation $\sum_i p_A(i)p_B(i)$; when two artists use their palettes differently the second is necessarily smaller by Cauchy–Schwarz. That null sits systematically high, and against it artists appear to agree *below* chance. We therefore use a paired null: shuffle B's cells and compare A against the shuffle, preserving B's own marginal while destroying its spatial arrangement, so both sides pair the same marginals.

Under the paired null, artists agree on 0.0977 of cells against 0.0859 expected (Figure 1) — 1.2 percentage points above chance, significant only because n is large, and a third of artist pairs (33.5%) fall *below* their own null. Per-cell agreement between independent artists carries almost no signal.

The finding is not specific to pixel art. We repeated the protocol on three independently designed emoji sets (Noto, Twemoji, OpenMoji; 64 emoji, 192 set pairs). Over all cells emoji redraws look twelvefold more consistent, $+0.1367$ against textures' $+0.0117$ — but two thirds of emoji cells are transparent background composited to white and agree trivially. Restricted to the shared foreground, with the null shuffled inside that mask, the gap is $+0.0154$ and a third of pairs again fall below their own null. Not excluding regions that agree trivially inflates this measurement ninefold.

3.2 Per-pixel objectives therefore measure marginal fit

If the conditional distribution over tiles is this wide, a per-cell loss against one artist's tile cannot separate a good sample from a bad one beyond how well it matches the marginal. We observed the consequence before we understood it: across three rounds of architecture changes to a masked-prediction model, validation accuracy rose steadily while the generated galleries did not visibly change.

3.3 VLM judges fail, and per-item agreement cannot detect it

We replayed a completed human blind study (72 pairs, §4) to three frontier judges under an identical protocol, asking each pair in both orders and discarding disagreements.

Order swapping earns its cost, but not for the reason usually given. None of the three judges is positionally biased on this content: asked 144, 128 and 144 times, they pick the first-presented image 54.2%, 51.6% and 48.6% of the time (p=0.36, 0.79, 0.80). What the swap exposes is instability.

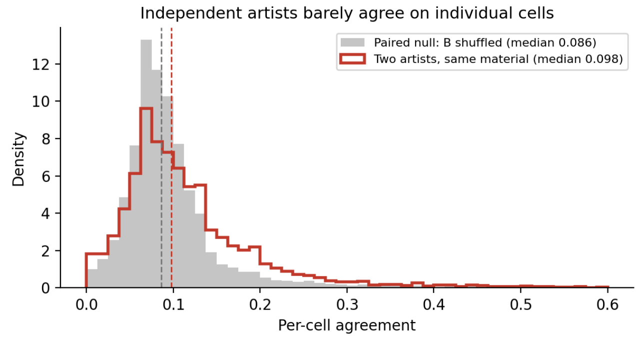

Figure 1: Independent artists agree on individual cells barely above a paired null. The null shuffles the second artist's own tile, preserving its marginal while destroying its arrangement, so both sides pair the same marginals.

Li et al. (2026) report order-swap consistency between 83.5% and 89.6%; here `claude-opus-5` is consistent on 72.2% of pairs ($[61, 82]$) and `gpt-5.6-sol` on 63.9% ($[52, 74]$), both intervals reaching no higher than that range's floor. Asked the same question twice, a judge changes its answer on a third of these pairs.

The results (Figure 2) separate into three failure modes (human rates below are recomputed over each judge's surviving pairs). `gemini-3.1-pro` compresses the human's 87% baseline win rate to 62% (p=0.14) and flattens the stratification that carried the study's main finding, reporting 37% where the human measured 6%. `gpt-5.6-sol` inverts it: 42% against the human's 78%, and 62% on the stratum where the human measured 4%. `claude-opus-5` preserves direction and significance (73%, p=0.008) and attenuates the stratification rather than erasing it (25% versus 33%).

The methodological point is panel (c). `gemini-3.1-pro` and `claude-opus-5` agree with the human on *exactly* the same fraction of items, 66.7%, and retain very different amounts of the effect: 62% against 73%, where the human measured 86%. Resampling the study's items, that ordering holds in 83% of draws and the significance threshold between the two in 54% (Appendix B), so the argument rests on how much of the effect a judge keeps, not on where p falls. Per-item agreement — the quantity this literature reports — is not evidence that a judge can be used.

Nor is stability. Ranked by order-swap consistency the order reverses: `gemini-3.1-pro` is the *most* self-consistent at 90.6%, clearing the top of the published range, while the two below it include the only judge we can use. Both cheap diagnostics — agreement with human labels, and agreement with oneself — are silent about what matters. A judge must be validated on whether it reproduces a conclusion humans have established.

A judge's bias direction determines what it can support. Because compression pulls any real difference toward 50%, a compressing judge manufactures null results. It can therefore support the existence of a large effect, as a lower bound, and can refute a null by finding significance in spite of compressing — but it cannot confirm a null. We apply that rule below rather than treating a validated judge as a general substitute for annotation.

4 A Trivial Baseline Is Enough

If per-pixel objectives measure the wrong thing (§3.1) and automatic judges cannot be trusted without validation (§3.3), then the only admissible arbiter left is a human blind comparison. Run under that arbiter, the ordering between a trivial baseline and a specialised model inverts relative to what every automatic quantity had reported.

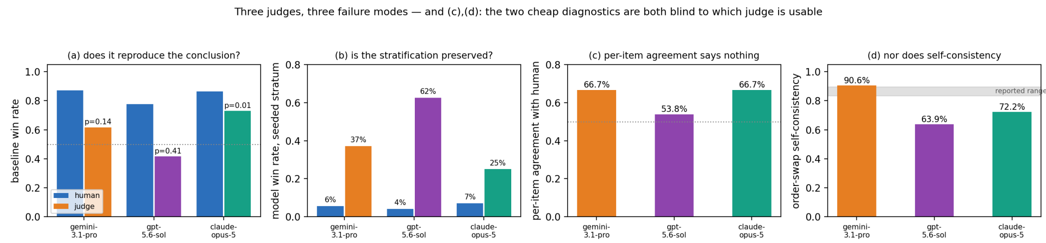

Figure 2: Three judges on the same human-labelled pairs. Panels (c) and (d) are the methodological point: the judge that destroys the conclusion and the one that preserves it have identical per-item agreement, and ranking instead by self-consistency under order swapping puts the judge that flattens the stratification on top, above even the range reported for web-development comparisons (shaded). Both cheap diagnostics are blind to which judge is usable.

4.1 The specialised model loses to the baseline

Our own three rounds of architecture work produced a mask-prediction model with hand-designed structural priors — brick period, border treatment, running-bond phase offset, per-material gating. We pre-registered a blind comparison against the trivial downsampling baseline: 72 pairs — 60 putting the model against the baseline, seven against artist tiles, and five attention checks, all passed — each item verified against the pipeline that produced it.

One pair was called a tie and dropped; the baseline won the remaining 59 49:10, 83%, $p{=}2.7 \times 10^{-7}$. Worse, the loss concentrates exactly where the priors act. On the seeded stratum — items where a structural prior actually fired — the model won $2/40$, 5%; on the unseeded control stratum it won $8/19$, 42% ($p{=}0.65$), close to a tie. The stratum difference is Fisher $p{=}0.001$, and its *direction is the opposite* of what we pre-registered: we had written that if the priors work, the seeded stratum should score higher. Win rate falls monotonically with structure score — 42%, 11% and 0% across equal-count tertiles ($\rho{=}-0.509$, $p{=}3.8 \times 10^{-5}$). An earlier comparison of the same model without priors lost to the baseline at 60%; adding the priors moved that to 83%. The priors did not fail to help — they actively hurt.

The structural ruler ranked them backwards. We had also built a structural-descriptor distance as a cheap quality proxy. It passed three validation criteria and was split-half stable, and it scored the seeded condition better than the unseeded one (0.93 vs 1.74) — the reverse of the human ordering (5% vs 42%). Its criteria established that the descriptor separates structured from unstructured content, which is a different question from whether a tile reads as the right material. Building the ruler succeeded; using it as a stand-in for quality was the error.

4.2 Against artists, the baseline is indistinguishable

The baseline beating our model would be uninteresting if the baseline were a low bar. It is not. Given a suitable high-resolution source, 24 comparisons (one called a tie and dropped) plus three attention checks put artist-drawn tiles against the downsampled baseline. Artists were preferred on $10/23 = 43\%$, $p{=}0.68$ — if anything slightly behind — in both strata (50% and 36%, Fisher $p{=}0.68$), with structure score not predicting the outcome ($\rho{=}-0.007$).

This was the third pre-registered outcome for that comparison, and the one we had least expected: the baseline is already at parity, leaving no room for a method to work in. It also overturns the premise the project started from — that low-resolution pixel art is an artist's redesign rather than a downsampling. An earlier attempt to test that premise with automatic features was declared falsified and then withdrawn because the ruler was blind to differences the eye could see; that escape is not available here.

What this comparison can and cannot support. With $n{=}24$ power against a true 65% artist preference is only 21%, so we ran the 60-pair extension through the validated judge: artists are preferred on $18/40 = 45\%$ ($p{=}0.64$, Jeffreys $[30\%, 60\%]$; attention checks $6/6$). Two arbiters that fail in different ways agree to within two points. A compressing judge cannot confirm a null, but because its compression is measured rather than assumed we can say what the result rules out: an artist preference of 65% would be expected to read as 60%, and the observed value is below chance (Appendix B).

5 The Real Bottleneck: Structural-Unit Convention

5.1 The failure is upstream of the downsampling kernel

Given a curated high-resolution source, the trivial baseline is already adequate (§4). The interesting case is a material that does not exist yet, where the source must be generated. There a 1024px render from a latent diffusion model (Rombach et al., 2022), in our case SDXL (Podell et al., 2024), looks structurally sound — a mossy brick wall with well-defined courses and mortar — yet every standard downsampling kernel turns it to mush at 16×16. We tried five (box, median, extremum, contrast-weighted, bimodal) and all five fail identically on the same renders, so the kernel is not the variable. Counting the structure explains why: the render packs roughly 25 brick courses into what must become 16 output cells. Improving the kernel cannot help, which is precisely where work on content-adaptive downscaling operates.

5.2 Artists hold a fixed number of units per tile

Measuring the dominant period of 5,979 artist tiles across three sizes gives median periods of 5.0, 10.0 and 8.5 pixels at 16, 32 and 64 — that is 3.20, 3.20 and 7.53 structural units per tile. Between 16 and 32 the unit count is unchanged: artists enlarge the units rather than adding more, and only at 64 do they add units. The period-to-side ratio is identical at 16 and 32 (0.312) and halves at 64 (0.133).

Which detector setting. Our detector searches up to 0.625 of the side; an earlier version stopped at 0.5, which by construction excludes two-course materials and found a period in $6/40$ stone-brick tiles against $36/40$ after the change. Under that earlier setting the same tiles give 4.0, 4.6 and 9.1 units, and the crop constant we use, 4.5, was fixed from those numbers and never re-tuned (§6). Both settings support the same qualitative claim; the production setting supports it more cleanly.

A second, independent measurement agrees. Across 125 artist-authored high-resolution sources with a detected directional period, a period spans 4.80 output cells once downsampled to 16 (IQR 3.84–7.17) — that is 3.33 units per tile, against the 3.20 the tiles give — with only 12% below the Nyquist limit of two cells per period. The same renders that fail above span 0.73 to 0.95 output cells per period, or 17 to 22 units per tile: five to six times more densely packed than the artist sources, and past the Nyquist limit by more than a factor of two. This is why curated sources need no intervention and generated ones do.

5.3 Fix and applicability

We crop the source to a centred window of side $4.5\times$ the detected dominant period — so that one period spans size$/4.5$ output cells — then downsample (Figure 3). Two conditions gate it: a period must be detected, and normalised anisotropy must reach 0.20. The anisotropy condition is necessary because period detection runs on row and column profiles and is meaningless for isotropic content. Across all 4,951 artist tiles, median normalised anisotropy is 0.287 where a directional period is detected and 0.063 where it is not; contrasting directional material classes (planks, brick, stone brick) against isotropic ones (foliage, gravel, metal blocks) separates them by more than an order of magnitude, 1.090 against 0.032. The threshold sits between the two groups either way.

Both conditions earn their place. Counting a firing on a directional class as a hit and on an isotropic class as a false trigger, the period-only rule at the earlier search range hits 80% and false-triggers 14%; widening the range raises both (98%, 20%); adding the anisotropy condition keeps the hits and removes the false triggers (96%, 4%). Repeating this on a held-out split assigned by material name rather than by inspection — 31 directional and 101 isotropic classes, 1,234 tiles, disjoint from the tuning classes — preserves the ordering: 71% at 39% for period-only, 83% at 18% for the gate. The

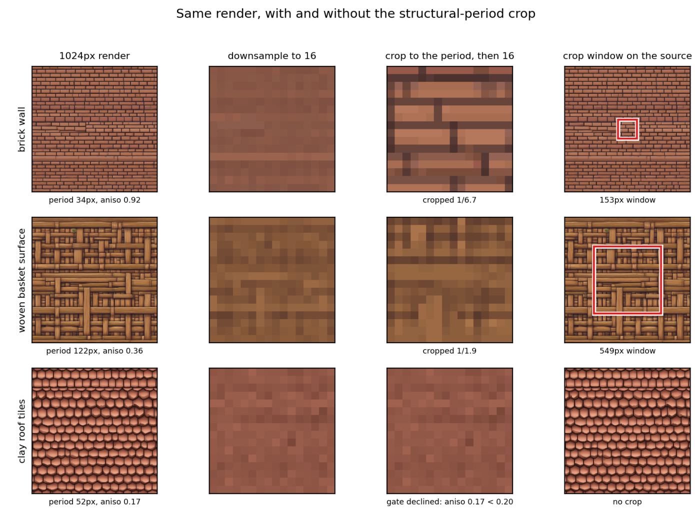

Figure 3: The same 1024px render, downsampled directly (second column) and after cropping to its detected period (third column); the red box shows the window taken. Brick courses and basket weave survive the crop and do not survive the direct reduction — and no downsampling kernel recovers them, because the structure is finer than the output grid before any kernel is chosen. The last row is a decline: clay tiles have a period but too little directionality ($0.17 < 0.20$), so the gate leaves the image alone and the two 16px panels are identical.

held-out false-trigger rate is an upper bound, since a name-based split files glass panes and ore veins as isotropic.

The gate fires on 45% of our prompts at 1024px — 58% on the disjoint set below, so the rate tracks the material mix — and the rest split in two. The conditions decline 38% (7% with no period, 31% below the anisotropy threshold). The other 17% pass both but cannot be cropped: 4.5 of their periods exceed the source, so the window is clipped to the whole image. That is a source-size limit with its own remedy — render larger, or take fewer periods — not a statement about which materials the method suits. The method therefore wins *within its applicability* and changes nothing elsewhere. Judged by `claude-opus-5` — the only judge that passed §3.3's validation — the crop is preferred on $31/35$ pairs, 89%, $p{=}3.5 \times 10^{-6}$, Jeffreys interval $[75\%, 96\%]$. Because that judge compresses, this is a lower bound.

Across sizes, and on materials we did not choose. The results above are at 16×16. Three screening protocols cover 16, 24 and 32. Two vary the judge and the render seed over one prompt set, giving $30/39$ (77%, $p{=}0.001$) and $38/43$ (88%, $p{=}2.5 \times 10^{-7}$) pooled across the three sizes. The third is a pre-registered replication on 60 materials disjoint from that set, with a third seed and the validated judge: $62/86$ (72%, $p{=}5.1 \times 10^{-5}$, Jeffreys $[62\%, 81\%]$), the gate firing on 58% of the new prompts. Each protocol leaves one tier individually non-significant — 32px in the first (57%, $n{=}14$) and 24px in the third (68%, $p{=}0.09$) — and the effect is smaller on materials we did not choose than on the set we did. All nine cells point the same way.

5.4 A PRE-REGISTERED FALSIFICATION SELECTS THE MECHANISM

The obvious reading of §5 is a sampling-limit account: structure finer than the output grid cannot be represented, so cropping restores what Nyquist removed. That account makes a prediction — as output resolution grows, more structure survives uncropped and cropping should matter less — and we pre-registered it over five output sizes with fixed criteria before running.

It failed (Figure 4). Across two independent protocols the trend is $\rho{=}-0.137$ ($p{=}0.277$) and $\rho{=}+0.044$ ($p{=}0.64$), and at 48px the crop wins 100% against 88% at 16px, triggering the pre-registered falsification condition. At 64px a 47px source period already spans 2.9 output cells, comfortably above Nyquist, and cropping still wins 79–90%.

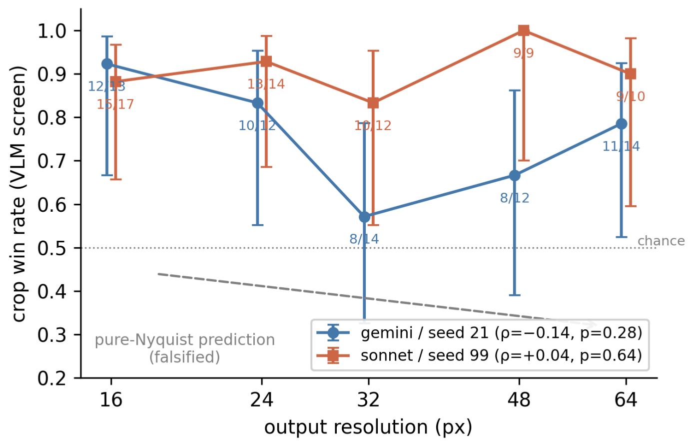

Figure 4: The pre-registered sampling-limit prediction, falsified. If cropping only undid aliasing, its benefit should decay as the output grid grows (dashed). Two independent protocols — different judge and different render seed — show no such decay, and cropping still wins at 64px where the source period is comfortably above Nyquist. Bars are Wilson intervals.

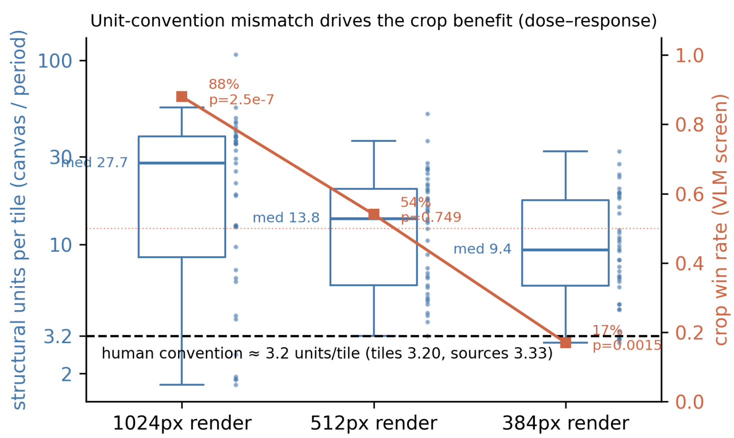

Figure 5: Left axis: the distribution of structural units per tile the generator draws at three render sizes, against the ∼3.2 artists hold (dashed; measured independently from 5,979 tiles and 125 sources). Right axis: how often cropping is preferred under each condition. As the generator's unit count approaches the artist convention, the crop stops helping and then hurts.

What survives is the convention account: the generator draws ∼25 units per tile *regardless of output size* while artists draw ∼3.2, an eight-fold mismatch that two independent measurements agree on (3.20 from tiles, 3.33 from sources). Because that mismatch does not depend on output resolution, this account predicts no gradient — which is what we observe. The failed prediction is what separates the two; had the gradient appeared, both would have been compatible with it.

5.5 Manipulating the unit count changes the benefit

We then tested the surviving account's own prediction. Prompt-level levers turned out to be inert: three variants failed a paired manipulation check on rendered period (p=0.38, 0.19, 1.00). A fourth, an explicit "few large blocks" modifier, had passed a manipulation check we wrote too loosely — it required only that the median move, with no significance test, so half a pixel of noise satisfied it. We report that as a void test rather than a result; under the criterion we then fixed, it fails as well, moving the period on 16/42 prompts (p=0.16). Render resolution does move it: on a separate 21-prompt probe the median unit count falls from 30.1 at 1024px to 8.8 at 384px (-70.9%, p=0.0066), clearing a manipulation criterion fixed in advance ($p < 0.05$ and a median change of at least 20%), while 512px did not clear it (-44.2%, p=0.081) and was excluded from the pre-registered test.

The prediction held. At 384px the crop is preferred on only 17% of pairs against 88% in the control, a drop of 71.7 points (p=6.3×10^{-9}), and the direction survives all three treatments of order-swap disagreements. Adding 512px as a descriptive midpoint yields a monotone dose–response (Figure 5): 27.7, 13.8 and 9.4 units map to 88%, 54% and 17%, with the midpoint sitting at no effect (p=0.749). These unit counts are from the crop study's own renders and so differ slightly from the probe's; both give the same ordering and the same verdict on the manipulation criterion.

The effect is condition-level, not per-image. Within a condition, per-image unit count does not predict the outcome ($\rho = +0.098$, p=0.316, n=106), because unit count there is confounded with material type. What is actionable is a setting-level rule — crop when rendering large, do not when rendering small — not a per-image gate. Our gate still fires on 48% of prompts at 384px where cropping hurts; we report that as a limitation rather than fitting a unit-count threshold the per-image data do not support.

6 Limitations

One annotator. Every human blind comparison we report — the 72-pair pre-registered study and the 24-pair artist comparison — was labelled by a single person, an author. Attention checks passed and items were verified against their pipelines, but inter-annotator reliability is unmeasured; the judge is a second arbiter rather than a second annotator. The labeller was not blind to the project's hypotheses, only to which condition produced each image.

A null we cannot confirm. Neither arbiter establishes equivalence. The human comparison is underpowered (21% against a true 65%), and the judge compresses, so its 45% bounds the effect using the compression measured on the validation set rather than proving its absence. "Indistinguishable" here means no large effect. The same asymmetry constrains every judge-mediated result: because our validated judge compresses effects, its numbers support lower bounds and can refute nulls, but cannot establish them (§3.3).

Conditional, not general, wins. The 89% preference is a within-applicability figure from a compressing judge, so it is a lower bound over a restricted domain, and the corresponding human study is prepared but unlabelled. We report no aggregate win rate over all prompts, because there is no defensible way to score the cases the method declines. The crop constant itself is inherited: it targets 4.5 units per tile, a figure measured before we widened the period detector, and the same tiles now give 3.20 (Appendix A).

The gate does not follow from the effect. The dose–response in §5 is condition-level, and within a condition per-image unit count does not predict the outcome (p=0.316). Our gate consequently misfires: at 384px it still triggers on 48% of prompts, where cropping hurts. Fitting a unit-count threshold would be unsupported by the per-image data, so the practical rule remains a setting-level one, and a per-image gate is open.

Where the gate declines, the baseline is weaker. Across 78 materials, tiles the gate declined carry a median realised luminance spread of 0.128 against 0.240 where it fired (Mann–Whitney p=1.1×10^{-4}; artists sit at 0.292) — systematically weaker but not uniformly flat, with 47% reaching the weakest tile the gate accepted. The obvious extension does not rescue them: artists draw these materials as per-cell colour rather than structural units, leaving no feature count for a crop to align to (Appendix B). Rescaling the realised spread to the artist median is preferred by the validated judge on $9/10$ pairs it answered consistently, but treats the symptom — tonal structure returns without the material becoming more recognisable. Half the task remains open, and it is the harder half.

One generator, two domains. The $\sim$25 units per tile is measured on SDXL renders; whether the mismatch takes the same magnitude, or the same form, in other text-to-image models is untested, and a model whose convention already matched would need no crop. The agreement finding replicates on emoji, but both domains are small-canvas raster redrawing by human authors, so we do not claim it for reference-scarce tasks in general.

Reproducibility Statement

The supplementary material contains the analysis code and the primary experiment records (raw judge outputs as JSON, with pre-registered decision criteria encoded in the evaluation scripts), sufficient to regenerate every figure and the statistical tests behind the paper's claims by running the released scripts directly on those records. Appendix A lists the results we withdrew, and why; Appendix B gives dataset construction, generation settings, the judging protocol and the statistical procedures. Hypotheses and decision rules were pre-registered as commits timestamped before the corresponding runs in the project's version history, which we will release together with the full experiment log upon publication.

AI Use Statement

LLM-based agents were used substantially throughout this project: for code implementation, experiment orchestration, statistical analysis, hypothesis refinement, and manuscript drafting. The human authors set the research direction, produced all human annotations, and reviewed the AI-generated code and text; every quantitative claim in the paper was verified against the raw experiment records by re-running the analysis scripts. Separately, commercial VLMs are *objects of study* in this paper: their use as judges is fully specified in the methodology, and their outputs are treated as measured data, not as verified claims.

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A WHAT WE RETRACTED

Every result below was written down as a finding, acted on, and later withdrawn when it did not survive checking. We list them because the alternative — reporting only what survived — would misrepresent how the conclusions in this paper were reached, and because several of the errors are of kinds that a reader could otherwise reproduce.

- **“$6\times$ closer to artists than the baseline”** was not a distance. The quantity was a difference of two group means, computed over four samples, and the same metric ranked the ablated model first. The corrected figure is $1.4\times$, and the metric was abandoned as a quality proxy: it passed three validation criteria and split-half stability while ordering conditions the *opposite* way from human preference (0.93 vs 1.74 against 5% vs 42%).
- **“Artists agree below chance”** came from the wrong null. Comparing artist A against a shuffle of A’s own cells has expectation $\sum_i p_A(i)^2$; the real comparison against B has expectation $\sum_i p_A(i)p_B(i)$, which is smaller by Cauchy–Schwarz whenever the two palettes are used differently. The null therefore sat above the signal and manufactured a negative effect. Under the paired null the gap is $+0.0117$ and positive (§3.1). The direction of the conclusion is unchanged; its strength is not.
- **A pre-registered resolution gradient** did not appear: $\rho = -0.137$ (p=0.277), non-monotonic, and $\rho = +0.044$ in an independent protocol. We report the falsification rather than the prediction, and it is what selects the surviving mechanism (§5.4).
- **A manipulation check we wrote too loosely** passed on noise. The criterion required only that a median move, with no significance test, so a half-pixel shift (p=0.95) satisfied it and the script printed “prediction confirmed”. We declare that test void rather than reporting it, and under the criterion we then fixed the same lever fails (p=0.16).
- **A second manipulation check that did not check the manipulation.** Testing whether rescaling a declined tile’s luminance spread to the artist median helps, we stretched the palette and measured the palette’s spread; it hit the target on $21/21$. The images barely moved — realised spread changed by under 0.05 on every one, and one tile went from twelve colours in use to one, because quantisation maps each cell to the nearest entry and spreading the entries only collapses more cells onto fewer of them. The judge discarded 14 of 21 pairs as order-inconsistent, which was itself the signal. We void that run. Stretching the image instead, and measuring the spread actually realised in the output, the manipulation lands and the pre-registered primary passes (§6). This is the same defect as the entry above, committed after writing that entry.
- **Cross-domain generality was inflated ninefold** by trivially agreeing regions. Measured over all cells, emoji redraws sit $+0.1367$ above the null against textures’ $+0.0117$; two thirds of emoji cells are transparent background composited to white. Restricted to shared foreground with the null shuffled inside the mask, the gap is $+0.0154$ — the same order as textures, which is the claim we make.
- **“Judges pick the left image 67% of the time”** did not survive measurement. The figure came from an early trial on a set where the human labels were themselves near chance, no per-item record was kept, and it disagreed in sign with the post-debiasing rates. Re-measured with both orders recorded, the three judges pick the first-presented image 54.2%, 51.6%

and 48.6% of the time (p=0.36, 0.79, 0.80). What the order swap does expose is instability, which is a different and more useful finding (§3.3).

- **The artist unit convention was measured with the wrong detector.** The script deriving it fixed the period search range at 0.5 of the side while the pipeline uses 0.625. Under the production setting the same tiles give 3.20 units per tile rather than 4.0–4.6, which the independent source measurement corroborates at 3.33. The crop constant 4.5 was fixed from the superseded numbers and never re-tuned (§6).
- **An off-by-one inflated a manipulation check.** A median helper returned `sorted(v)[len(v)//2]`, the upper median, which at even n turned 30.1/16.8/8.8 into 32.5/17.4/9.3 and p=0.0066 into p=0.0062. The pre-registered verdicts are unaffected. We swept the codebase for the pattern and fixed the remaining instances, none of which had reached a reported number.

Two of these — the misspecified null and the loose manipulation check — share a shape worth naming: a comparison whose reference point was chosen without asking what it would produce under the null hypothesis. Both passed casual inspection and produced the result we expected.

B Experimental Detail

Tiles. Texture packs are collected from the Minetest ContentDB. Packs whose licences forbid derivative use in training are excluded and were never used for that purpose. Each tile is quantised to at most k=16 palette entries by median-cut on RGB, keeping only the entries the tile actually uses (the median tile natively carries 21 distinct colours; 2,933 of the 5,979 tiles across all sizes use the full 16). Materials are matched across packs by file name, which is stable in this ecosystem. The agreement study uses all 16×16 tiles: 4,951 tiles, 517 materials, 64 packs, 50,753 same-material pairs. Held-out packs for the modelling work are listed in the dataset file and were fixed before any model was trained.

Emoji. Noto (Apache/OFL), Twemoji (CC BY 4.0) and OpenMoji (CC BY-SA 4.0), 64 single-codepoint emoji present in all three sets, composited onto white, box-reduced to 16×16 and quantised to 12 luminance levels — the same pipeline as tiles, with luminance standing in for the palette index. The foreground mask is the intersection of the two sets' $\alpha > 0.5$ regions, and the null shuffles cells inside that mask.

Generation. Every experiment in this paper uses SDXL base 1.0 with no adapter, fp16, 28 denoising steps, at 1024×1024 unless stated. Prompts are wrapped as *"pixel art, {material}, top-down seamless tileable game texture, flat lighting, no shadows, orthographic, chunky large pixels"* with a negative prompt of *"perspective, 3d render, vignette, watermark, text, border, blurry"*. The seed for the ith prompt is base + i; the main crop study uses base 21 and its independent replication 99. Our delivery tool optionally loads a pixel-art LoRA, but no reported result does.

Period and anisotropy. The dominant period is the first significant peak of the autocorrelation of row and column intensity profiles, searched from 8px up to 0.625 of the side on sources and from 2px on 16×16 tiles. Anisotropy is $|h - v|$ over the mean of h and v, where h and v are mean absolute horizontal and vertical gradients; it is scale free and unaffected by overall contrast. The crop takes a centred window of side $4.5 \times$ period, so one period spans size/4.5 output pixels, clipped to at least 0.08 of the source side.

Judges. Each pair is presented as two images and the judge is asked which looks more like the named material — for the crop screening, which also looks more like a usable game texture — answering only A or B, at temperature 0 with a cap of 8 output tokens. Every pair is asked in both orders and pairs answered inconsistently are discarded; reported rates are over the survivors. Attention-check items pair a clean tile against a blurred one. At temperature 0 the judges are deterministic — two runs agreed on 35/35 jointly answered pairs — so a same-seed rerun is not an independent replication, and our replications vary both the judge and the render seed.

Human studies. Items are shown one pair at a time in a static HTML page with left/right assignment randomised per item and the order of items shuffled. Attention checks are interleaved at a rate between about one in twenty in the largest study and one in nine in the smallest. Both studies were labelled by one author, blind to which condition produced each image but not to the hypotheses; response times were recorded and no item was answered implausibly fast (median 1.4 s in the artist comparison, with two items under 0.8 s).

Statistics. Binomial tests are exact, computed with `math.comb`; interval estimates are Jeffreys; two-condition comparisons use Fisher's exact test by summation over the hypergeometric tail; monotonicity uses Spearman's ρ with tie-corrected ranks. No multiplicity correction is applied anywhere: each confirmatory verdict rests on a single pre-registered primary comparison whose decision rule was fixed in advance, and every other p-value is exploratory and reported uncorrected. The implementations carry no SciPy dependency, because the environment the experiments ran in did not have it and three earlier runs lost their summary statistics to that import after the data had already been written.

Calibrating what a compressing judge rules out. A judge that compresses cannot establish a null, but a measured compression factor turns its answer into a bound. On the validation set `claude-opus-5` under this protocol returned 73% where the human measured 86%, retaining $(0.73 - 0.5)/(0.86 - 0.5) = 0.64$ of the distance from chance. Applying that factor, a true artist preference of 65% in §4 would be expected to read as 60%, 70% as 63% and 75% as 66%. The judge returned 45%, with a Jeffreys interval reaching only 60%. This does not show the two are equivalent; it shows an advantage of the size that would change the paper's conclusion is inconsistent with what the judge returned. The factor comes from one validation study and is applied to a different comparison, so it is an argument about magnitude, not a calibrated estimator.

How much of §3.3 rests on 72 items. Every judge verdict there is validated against one blind study, so we resampled its items with replacement (B=4000, each judge over the pairs it answered; the three answered different subsets, overlapping on only 18). The direction findings are firm: `gpt-5.6-sol` inverts in 86% of draws and preserves in 0.1%, `claude-opus-5` inverts in 0.2%. The ordering of how much effect each retains — 73%, 62%, 42% against the human's 86% — holds in 83% of draws. What is *not* firm is the significance threshold: `claude-opus-5` clears $p < 0.05$ in 78% of draws and `gemini-3.1-pro` in 31%, so the statement that one is significant and the other is not holds in 54%. Nor is the stratification gap, which the human puts at -27pp: `claude-opus-5` gives -8pp with a resampling interval of $[-45, +27]$. Section 3.3 is written against the two robust legs and says so; `judge_bootstrap.py` reproduces all of these.

Why a crop cannot help the declined materials. For the tiles the gate declines, we asked whether the same unit convention applies in a non-directional form. The 1D profile estimator in the pipeline is unsuitable here — isotropic speckle cancels in row and column means — so we used the azimuthal average of the 2D autocorrelation instead. On 84% of those tiles it is already below half at a radius of one cell, and relaxing the threshold to 0.2 moves the median only to two. There are no features to count, so a crop has nothing to align to. This is also why cropping granular materials by a characteristic scale looked worse than leaving them alone, an observation recorded in the code long before it had an explanation. The criterion for this test was committed before it ran.

The generation-side form of the lever fails here too. Rendering these materials at 384px instead of 1024px before quantisation raises the realised luminance spread on every declined material (median $+0.57$, $18/18$, sign test $p = 8 \times 10^{-6}$), overshooting the artist median of 0.29, yet the validated judge preferred the 384 version on only 2 of 13 consistent pairs (15%, $p = 0.02$) — significantly the reverse. Criteria were committed before the run. Brightness range returns, but recognisability does not follow: for this half of the materials, resolution is not the missing lever at any stage of the pipeline.